

Parameters

Contents

Contents	1
Categories of Parameters	5
Knife and Drill Speeds Parameters	6
Idle Knife Speed	6
Dry Haul/Minimum Knife Speed	6
Cut/Maximum Knife Speed	6
Sharpen Knife Speed	6
Cut Speed=Maximum Knife Speed	6
Examples.....	6
Heavy Duty Drill Speed.....	6
Knife Spd Sensor: Volts per KRPM.....	6
Knife Speed Checking	6
Sharpen Parameters	7
Sharpen Frequency	7
Cut Distance Limit.....	7
Sharpening Time.....	7
Sharpen Position #1	7
Sharpen Position #2	7
Number of Sides to Sharpen	7
Sharpen Every n Pieces	7
Sharpen Only When Knife is Up	7
Knife Wear Type	7
Knife Wear Limit.....	8
Sharpens Until Overcut Update	8
Sharpen New Knife n Times.....	8
Num sharpens until angle adjust.....	8
Number of Sharpen Cycles	8
Cutting Parameters	9
Active Beams.....	9
Dual Beam Cutter	9
Slit Notch Backtracking.....	9
Optimal Backup Distance	9
Advance Before Plunge Distance	9
Corner Advance Distance	9
Overcut Distance	9
Corner Overcut Distance	9
Lift & Plunge Corner Angle	9
Maximum Segment Length	10
M17 (0 Heel Cut) Advance.....	10
M19 (Slit Notch) Advance.....	10

Contents

M25 Slowdown Percentage.....	10
Miter Angle Range.....	10
Corner Miter Distance.....	10
C Axis Yaw Angle	10
C Axis Max Angle	10
Vacuum System Parameters	11
Vacuum Zoning.....	11
Vacuum Zone Overlap.....	11
Vacuum Error Reporting.....	11
Vacuum Control Type.....	11
Vacuum Level — Bite	11
Vacuum Level — Cut	11
Vacuum Level — Idle.....	12
X Home Zone1 Offset.....	12
Vacuum Low Error Level	12
Vacuum Low Caution Level	12
Vacuum Control — Pattern Length	12
Vacuum Control — Zero Offset	12
Vacuum Control — Vacuum Gain.....	12
Number of Vacuum Zones	12
Bite Vacuum Threshold	12
Time Until ESM.....	12
Energy Savings Mode	12
Num of Vac Zones Closed for Bite.....	12
Vacuum Level Valve on Inlet	12
Ignore Codes Parameters	13
Ignore Auto Origin (M70) Codes.....	13
Ignore Aux Drill (M44) Codes.....	13
Ignore Display Msg (M20) Codes.....	13
Ignore Feedrate (F) Codes.....	13
Ignore Label Cycle (M31) Codes.....	13
Ignore Lift & Plunge (M46) Codes.....	13
Ignore Main Drill (M43) Codes	13
Ignore Move Conveyor (M69) Codes	13
Ignore Overcut Off (M18) Codes.....	13
Ignore OV, AP Off (M19) Codes.....	13
Ignore Pen (D1+ D2) Codes.....	13
Ignore Sharpen (M42) Codes	13
Ignore Slowdown Codes	14
Bristle Cleaner Parameters	14
Bristl Cleanr (0=N 1=Motr 2=Pneum).....	14
Slat Advance Time.....	14
Cleaner Head Sweep Time.....	14
Cleaner Head Retract/Plunge Time.....	14
Bristle Cleaner Sweeps per Slat	14
Number of Table Slats	14
One-Way Bristle Cleaning	14

Contents

Knife and Path Intelligence Parameters	15	Convey Last Bite Off Table	20
KI: Gain	15	Bite Clip Start Position.....	21
KI: Deflection Angle	15	Bite Clip End Position.....	21
KI: Max Turn Angle	15	Bite Park Offset.....	21
KI Spring Factor	15	Insert M23 Edge Clip Code.....	21
KI: Error Detection	15	Bite Veeroff.....	21
KI: On/Off	15	Marker End Park (0=None 1= -Y 2= +Y).....	21
KI: Option	15	Marker End Park Position X	21
KI: Fail Time	15	Marker End Park Position Y	21
KI: Corner Disable Time	15	Maximum Bite Adjustment.....	21
Only Allow Cutting with KI On	15	Bite Park Location.....	21
Path Intelligence Option	16	Cut Small Pieces First	22
PI FR Control — XY Force.....	16	Max Area of Small Piece	22
PI FR Slowdown — Low Vac	16	Recalculate Small Pieces.....	22
PI FR Faster — Shp Knife.....	16	Bite Overlap	22
PI Optimum Force	16	Conveyor Table Settle Time	22
PI Feedrate Max Increase	16	Continuous Cutting	22
PI Feedrate Minimum	16	Continuous Cutting Installed.....	23
PI Feedrate Maximum	16	Options and Modes	24
PI Feedrate Change Ratio	16	Use Edge Finders	24
PI Velocity to +FR.....	16	Edge Finder Auto-Alignment.....	24
PI Velocity to +FR & Rbld.....	16	Log Edge Finder Info	24
PI Velocity to -FR.....	16	Auto-Scale Y	24
PI Velocity to -FR & Rbld.....	16	Max Auto-Scale Y Change	24
PI Force to Stop with Warning	16	Dry Run Tool is Light.....	24
PI Sharp Knife Full Speedup Dst.....	16	Clear Origin on Cutfile Close	24
PI Sharp Knife Prtl Speedup Dst	17	Installed Edge Finders.....	24
PI Sharp Knife Speedup Amount.....	17	Saving of Parameter Changes	24
PI Vacuum Low Slowdown Factor.....	17	Single Step	25
Knife Break Force.....	17	Marker Flip	25
Knife Break Risk Point.....	17	Marker Flip Top	25
Data Format/Scaling and RSO	18	Marker Stacking Reorder	25
X Data Scale Factor	18	Marker Stacking Reorder Channel.....	25
Y Data Scale Factor	18	Language Support	25
Data Format	18	Kanji Character Set	25
Marker Size: Use Auto Calculate	19	Check Origin	25
Marker Size: Use M20 Code	19	Verify Keyboard Entry.....	26
Right Side Initialize	19	Block Delete.....	26
AAMA Data Conversion	19	Cut/Plot Mode	26
Center Line Calibration	19	Dry Run.....	26
Bite Parameters.....	20	Metric Display.....	26
Allow Piece Reorder.....	20	Delete Cut File After Cutting.....	26
Run Time Biting	20	Re-Sealer Option.....	26
Variable Length Biting	20	Resealer Up For Continuous Cut	26
Minimum Bite Length	20	Preprocess Drills	26
Maximum Bite Length.....	20	Center Line Origin Center Point	26
Pause Before Convey.....	20	Center Line Origin Edge Points	27
Pause Between Bites.....	20	Transfer Markers When Selected.....	27

Contents

Y Axis Edge Align (0=N 1=+Y 2=-Y)	27	Offset in X — Pen.....	31
Bar Code Reader Option.....	27	Offset in Y — Pen.....	32
Cutter Info Database Option.....	27	Ramp Down Dist — Auto Bite	32
Mirror Image and Clipping Window	28	Offset in X — Drill 2	32
Mirroring Option	28	Offset in Y — Drill 2	32
Mirror Image.....	28	Offset X — Camera	32
Y-Axis Mirror Image.....	28	Offset Y — Camera	32
Y-Axis Mirror Reflection Line	28	Offset Edge Sensors X.....	32
Clipping Window Option	28	Offset Bottom Edge Sensor Y.....	32
Clipping Low Boundary	28	Offset Top Edge Sensor Y	32
Clipping High Boundary.....	28	Max Extension -Y Edge Finder.....	32
Clipping Low Boundary (Top Mrk)	28	Max Extension +Y Edge Finder.....	32
Clipping High Boundary (Top Mrk).....	28	Cut File Preprocessor Parameters.....	33
OPSTOP and Report Options	29	Length of Reverse Segment	33
Camera Extension Distance	29	Edge Clipping Angle.....	33
Use AccuMatch	29	Knife Width.....	33
OPSTOP Section Processing	29	Gap to Ignore	33
OPSTOP	29	Distance to Corner	33
Preset OPSTOP.....	29	Path Overlap.....	33
Regional OPSTOP	29	Corner Angle.....	33
OPSTOP Slow Slew	29	Transfer Angle	33
Cutter Info Database Output.....	29	Max Length to Ignore.....	33
Selvage Cut/Contiguous Cut	30	Max Distance with Knife Down	33
Selv/Web (0=Off 1= Selv 2 = Web&Sel).....	30	Message Stop	33
Selvage Cut Position.....	30	Max # Points Per Piece	33
Selvage Cut Size – Bottom	30	Overlap Distance.....	33
Selvage Cut Size – Top.....	30	Smallest Piece	33
Minimum Web Cut Size	30	Percent of Piece.....	33
Intermarker Gap.....	30	Queue Manager Parameters.....	34
Minimum Size of First Bite	30	Auto Start Jobs in Queue	34
Contiguous Cut	30	Del Job From Queue After Cutting.....	34
System Setup Parameters	31	Del Queue After Cutting.....	34
Beam Square Adjustment	31	Read Queue Name Bar Codes.....	34
Beam Squared.....	31	Security Parameters.....	34
Bite Error Adjustment.....	31	User Parameter Table - Selecting.....	34
Maximum Cut Velocity	31	User Parameter Table -Editing.....	34
Max Cut Acceleration.....	31	Field Service Level Categories and Parameters	35
C Axis Sharpen Speed	31	Cutter Type	35
X-Axis Stepsize	31	Dual Beam.....	35
Y-Axis Stepsize	31	AccuMatch Installed.....	35
Usable Table Distance in X	31	Table Length (0=Standard,1=Long)	35
Usable Table Distance in Y	31	Beam Width.....	35
C Home to System 0 Degrees	31	Right Side Control.....	35
Offset in X — Knife	31	PC Serial Communication.....	36
Offset in Y — Knife	31	MCC Communications Port	36
Offset in X — Drill 1	31	MCC Comm Port Beam 2.....	36
Offset in Y — Drill 1	31	Bar Code Reader Com Port	36

Contents

Dry Haul Speeds	36	Conveyor and CTOT	41
Projector Output Port	36	Velocity — Cnvyr Jog	41
Maximum Dry Haul Velocity	36	Velocity — Cnvyr Man Bite	41
Dry Haul Acceleration	36	Velocity — Cnvyr Auto Bite	41
C-Axis	37	Conveyor Ramp Down Time	41
C-Axis delay	37	CTOT — Stop Distance	41
C-Axis cut velocity	37	Max Allowable Conveyor Friction	41
C-Axis max velocity	37	TONT — Start Delay	41
C-Axis Acceleration	37	TONT — Stop Distance	41
C Axis Settle Time	37	CTOT2 — Start Delay	41
XY Axes	37	CTOT2 — Stop Distance	41
X Home to System 0	37	Bite Encoder Step (beam)	41
Y Home to System 0	37	Conveyor Encoder LOC (0=B 1=T)	41
XY-Axes delay	37	Bite Encoder Step (table)	41
Cutfile Percentage Primary Beam	37	Velocity — Cnvyr Cut Bite	41
Limit Switches	38	Variable Speed CTOT	41
Lm Switch Status — Bite Sensor	38	Software Parameters	42
Lm Switch Status — Drill 1	38	Event Logging	42
Lm Switch Status — Knife	38	Data Logging Mask	42
Lm Switch Status — Pen	38	Milliseconds between samples	42
Lm Switch Status — Sharpen Arm	38	Log Writes to VARSAVE	42
Lm Switch Status — Drill 2	38	Log OPSTOP Debug Info	42
Limit Switch Status — Edge Sensor	38	Log Origin Debug Info	42
Drill	39	Log CIDB Debug Info	42
Drill Idle Time Before Shutoff	39	Index	43
Drill on only while selected	39		
Servo (DSP) Coefficients	39		
C Max Follow Error	39		
Beam Bite Gain	39		
Time Considerations	40		
Max time: To move up — bite sns	40		
Max time: To move down — bite sns	40		
Max time: To move up — drill 1	40		
Max time: To move down — drill 1	40		
Max time: To move up — knife	40		
Max time: To move down — knife	40		
Max time: To move up — pen	40		
Max time: To move down — pen	40		
Max time: Up to speed — knife	40		
Max time: Up to speed — drill	40		
Max time: To turn off — drill	40		
Max time: To turn on — vacuum	40		
Max time: To turn on — drives	40		
Max time: Idle in TrimCut Mode	40		
Max time: Idle in ManBite Mode	40		
Max Time: Vacuum Drop	40		
Max Time for Auto Bite	40		
Max Time: Engage Edge Sensor	40		

Categories of Parameters

Parameters are settings that are included in the setup for your cutter. Some parameters turn a feature on or off. Other parameters define a value within an allowable range. For example, use the Clipping Window Option parameter to turn this feature ON and then set current values using Clipping Boundary parameters.

Note: This chapter explains available parameters for all GERBERcutters. However, only parameters for your cutter type may display in the Parameter Manager.

Categories group parameters that relate to a cutter feature. The following parameter categories can be accessed at the User and Supervisor levels. Additional parameter categories can be accessed at the **Field Service** and **Software** levels.

Category /Parameters Used To

Frequently Used Parameters —

Provide quick access to frequently-used parameters

Knife and Drill Speeds —

Control knife and drill speed for specific cutting conditions.

Sharpen Parameters —

Control how and when the knife is sharpened.

Define how the cutter determines wear on the knife blade and when the knife should be replaced.

Cutting Parameters —

Control advance and overcut distances and limits.

Control cut angle and slowdown percentage for cutting operations.

Vacuum System —

Control the vacuum options.

Ignore Codes —

Tell the cutter to ignore or respond to certain marker data codes.

Bristle Cleaner —

Control times and number of sweeps for automatic cleaning.

Knife and Path Intelligence —

Control how Knife Intelligence compensates for cutting forces on knife.

Set up Path Intelligence to automatically adjust feed rate for the best cut quality and throughput.

Categories of Parameters

Category /Parameters Used To

Data Format/Scaling and RSO —

Define the format of cut file data or AAMA data conversion.

Define X and Y-axis scaling.

Define marker size.

Initialize RSO cutters.

Bite Parameters —

Define how the cutter handles bites.

Define end park position.

Options and Modes —

Turn on/off various GERBERcutter features.

Mirror Image and Clipping Window —

Turn on the mirroring options which rotate the marker before cutting.

Turn on the clipping window feature and define boundaries.

OPSTOP and Report Options —

Control how the cutter responds to OPSTOP codes.

Define Cutter Info Database reporting.

Selvage Cut/Contiguous Cut —

Control Selvage/Web cutting for vertical line cuts through the selvage or the selvage and web.

Control processing of consecutive markers.

System Setup Parameters —

Set maximum beam velocities for cutting.

Cut File Preprocessor —

Determine the parameters used when preprocessing markers.

Queue Manager —

Determine how Queue Manager handles cut data files.

Security —

Control the ability to change or edit Cutter and Notch parameter tables when opening an AccuMark marker for cut processing.

Knife and Drill Speeds Parameters

Knife and Drill Speeds Parameters

Parameters in this category control knife and drill speeds for specific cutting conditions.

Idle Knife Speed

Sets the speed at which the knife reciprocates whenever the system is in manual mode (idle).

Dry Haul/Minimum Knife Speed

Sets the speed at which the knife reciprocates during dry haul in auto mode. This is the minimum knife speed (feed rate 1) during cutting.

Cut/Maximum Knife Speed

Sets the maximum knife speed during cutting. The knife reciprocates between the minimum and maximum settings as it cuts material.

Sharpen Knife Speed

Sets the speed at which the knife reciprocates during sharpening.

Cut Speed=Maximum Knife Speed

Sets the value on the feed rate dial/slider at which the maximum knife speed (RPM) is reached. As you increase the feed rate, knife speed increases from minimum to maximum speed in equal increments. Once maximum knife speed is reached, the knife speed will stay at the maximum speed.

If your operation requires a wide range of cutting speeds, you might set cut speed=15 or feed rate equal to 100%. This allows the most increments before the maximum knife speed is reached.

If your operation requires that you reach maximum cutting speed quickly, you might set cut speed=6 or feed rate equal to 40%. This example would allow fewer speed increments before the maximum knife speed is reached.

Examples

GERBERcutter	GT 3250	GTxL
Maximum Cut Velocity (ipm)	1800	1800
Use to set feed rate:	Feed Rate switch on control panel	Feed Rate Slider in C-200MT
Feed rate range	0-15	0-100%
Setting = ipm	0 = 0 ipm 1 = 120 ipm 15 = 1800 ipm	0% = 0 ipm 10% = 180 ipm 100% — 1800 ipm

Heavy Duty Drill Speed

Sets the speed of the electric drill. The electric drill is standard on most GERBERcutters and available as an heavy-duty drill option for GT 3250 or S-3200 cutters.

Knife Spd Sensor: Volts per KRPM

Sets the factor used to change voltage to revolutions in order to calibrate knife speed. This factor depends upon the stroke size of the cutting head. The result of this calibration displays in the Knife Speed gauge in the Main Cutter Control screen.

Knife Speed Checking

ON — Checks for knife oscillation before allowing cutting.

OFF — Turns off this feature.

Sharpen Parameters

Parameters in this category:

- Control how and when the knife is sharpened
- Define how the cutter determines wear on the knife blade and when the knife should be replaced

Sharpen Frequency

Tells the system to automatically sharpen the knife whenever it has cut this distance in automatic cutting. Sharpening frequency depends upon your application. The highest frequency combined with the lowest Number of Sharpen Cycles results in quality cutting and greatest knife life.

Note: Typing a value of 0 (zero) disables automatic sharpening.

Cut Distance Limit

Sets the maximum distance (centimeters or inches) the knife can cut before you must replace it. If Knife Wear Type is set to DISTANCE, the system uses this setting to define the life of a knife blade; otherwise, it uses Sharpen Every n Pieces.

Knife life limit reached is the message displayed on the monitor when the knife has traveled the distance you set in this parameter. When you see this message, install a new knife.

Note: After replacing the knife, use the New Knife and Sharpen commands in the Cutter menu to reset Knife Wear logic.

Sharpening Time

Sets the amount of time that the knife blade stays against the sharpening stones for each sharpening cycle; for GT 3250 or S-3200 only.

Sharpen Position #1

Sets the C-axis angle for sharpening the first side of the knife blade; for GT 3250 OR S-3200only.

Sharpen Parameters

Sharpen Position #2

Sets the C-axis angle for sharpening the second side of the knife blade; for GT 3250 OR S-3200only.

Number of Sides to Sharpen

Sets the number of knife blade sides sharpened during each auto mode sharpening cycle; for GT 3250 OR S-3200only. The choices are: 1 or 2

Note: Manual mode always sharpens both sides of the knife blade.

Sharpen Every n Pieces

Automatically sharpens the knife every “n” pieces, where “n” is a number from 1 to 9. Use when pieces in markers are similar enough in size that distance can be estimated accurately.

If “n” is 0, the system uses Sharpen Frequency instead of this parameter to sharpen the knife.

Note: When using Sharpen Every n Pieces to sharpen the knife, set Sharpen Frequency to 999 to prevent back-to-back sharpening.

Sharpen Only When Knife is Up

ON — Postpones next sharpen cycle until the knife lifts out of the material for another reason (for example, a Lift & Plunge or a Dry Haul). Use when there are many lift & plunges in a marker. Use with the Sharpen Frequency parameter.

OFF — Does not postpone sharpen cycle.

Knife Wear Type

Determines how the system defines knife wear:

CYCLES (used for most applications)

DISTANCE (aerospace industry)

Sharpen Parameters

Knife Wear Limit

Sets the maximum number of sharpening cycles allowed before you need to replace the knife blade. If Knife Wear Type is set to CYCLES, the system uses this setting to define the life of a knife blade; otherwise, it uses Cut Distance Limit.

Knife life limit reached is the message displayed on the monitor when the knife has reached the sharpening limit. When you see this message, change the knife.

Sharpens Until Overcut Update

If Knife Wear Type is set to CYCLES, the C-200MT uses this parameter to know when to update Knife Wear compensation. This setting is the number of sharpening cycles before the C-200MT updates Knife Wear compensation. Knife Wear Logic allows 0.2 mm (0.008 in.) of blade wear at each update. The system automatically adjusts Overcut and Corner Overcut.

Sharpen New Knife n Times

Sets the number of times the system sharpens a new knife blade when you use the New Knife command in the Cutter menu.

Num sharpens until angle adjust

Sets the number of sharpen cycles until the system automatically adjusts the Sharpen Position angles by 0.250 degrees (maximum 4 degrees). Used for GT 3250 OR S-3200 only to maintain proper sharpening (based on the fact that knife width decreases due to normal sharpening wear).

Number of Sharpen Cycles

Sets the number of rotations, clockwise and counterclockwise, during a knife sharpening interval; not used on GTXL, GT 3250, OR S-3200.

Cutting Parameters

Parameters in this category:

- Control advance and overcut distances and limits.
- Control cut angle and slowdown percentage for cutting operations.
- Adjust for new or used knife.

Active Beams

Identifies which beam(s) are being used on a cutter with the Dual Beam option.

Choices are: PRIMARY, SECONDARY, or BOTH

Dual Beam Cutter

Identifies whether or not the GERBERcutter has the Dual Beam option. This option, which utilizes two cutting heads each on a separate beam, is only available for S-91 ZVC cutters.

YES — Turns on this option.

Slit Notch Backtracking

ON — Turns on backtracking when cutting slit notches. Backtracking on the piece perimeter by a specific amount (optimal backup distance) prevents incomplete cutting which might result due to material movement while cutting the notch.

OFF — Turns off this feature.

Optimal Backup Distance

Sets the amount the cutter backtracks on the piece perimeter after making a slit notch. The Slit Notch Backtracking parameter must be turned ON.

Advance Before Plunge Distance

Sets the distance the knife advances before plunging into the material at the start of a piece or line segment. This sets the amount of heel cut left outside the piece during the first knife plunge. Heel cut is made by the thickest part (heel) of the knife; for thicker knives, set this value larger.

CAUTION: Too much heel cut can damage an adjacent piece. Too little heel cut can cause the piece not to be fully cut away from the scrap material.

Corner Advance Distance

Controls the advance at a corner where the knife lifts from the material, turns to a new heading, plunges into the material, and resumes cutting.

Note: This value is usually the same as Advance Before Plunge Distance, unless the shapes and angles for pieces in a marker are very different.

Overcut Distance

Sets the extra distance the knife cuts when it stops cutting a piece or internal line, but not a Lift & Plunge corner. This ensures that the piece is cut cleanly from the web.

Note: Sharpen Until Overcut Update changes this value.

Corner Overcut Distance

Sets the amount of extra material cut at a Lift & Plunge corner only.

Lift & Plunge Corner Angle

Sets the knife angle that makes the system execute a lift and plunge corner cut. If the angle at the corner of a piece is equal to or less than the Lift & Plunge Corner Angle, the knife lifts out of the material, turns, and plunges back into the material; if wider, the knife cuts the corner without lifting out of the material.

Cutting Parameters

Maximum Segment Length

Sets the maximum move (distance) between data points allowed by the system. This value must be equal to or greater than the value specified in the CAD marking system. For example, AM-5 MPLOT routine, AccuMark™ Cutter parameter table, or MicroMark™ NC Cut table.

Note: If segments greater than 100 inches (254 cm) are required, digitize intermediate points.

M17 (0 Heel Cut) Advance

Causes the cutter to make an advance for zero heel cut, such as cutting web. Advance returns to normal after each M17 code.

M19 (Slit Notch) Advance

Causes the cutter to make an advance for cuts such as slit notches.

Note: There is no lift and plunge on slit notches.

M25 Slowdown Percentage

Sets the percentage of slowdown caused by M25 code. This improves cutting production by running optimum speed except at curves.

Example: A value of 75 means the cutter slows down to 75% of cut speed.

Miter Angle Range

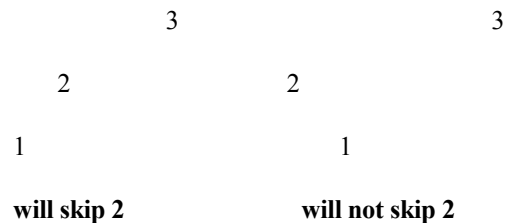
Defines the range of angles at which corner mitering will take effect. The setting is relative to 90 degrees. For example, if this value is set to 20, mitering will occur at all corners cut with the knife down at angles from 70 to 110 degrees.

Corner Miter Distance

Determines the distance away from the original corner point where the miter points are inserted to replace the corner point. Setting this parameter to 0 (default) disables the feature.

Advance Ignore Angle

Sets an angle the system uses as reference to minimize or eliminate knife advances that can cause heel cuts in pieces with small line segments. The system checks the distance between point 1 and 2 (shown below); if this is less than the size of the knife, the system calculates the angle between point 1 and point 3 and compares to the Advance Ignore Angle. If less than this angle, the system skips point 2 and cuts between point 1 and point 3, trying to minimize the overcut and undercut.



C Axis Yaw Angle

Changes the yaw angle of the knife up to 2 degrees in either direction without altering home position or sharpen angle. This is helpful when resistance is encountered during cutting.

Note: Cutting direction and yaw angle affect piece size. For example, cutting clockwise with a — Yaw angle yields a piece that is slightly larger.

Setting Sharpen Angle GT 3250 or S-3200

When setting the sharpen angle for GT 3250 OR S-3200 cutter, first zero this angle using the C Axis Angle diagnostic, set the Sharpen Position #1 and #2 parameters, and then use this parameter to make any necessary adjustments.

C Axis Max Angle

Sets the angle that determines whether the system stops for a brief instant when cut direction changes or changes direction without stopping at all. Useful when cutting without lift and plunge; for example, small angles at bottom of a pocket.

Vacuum System Parameters

Parameters in this category can be used to control vacuum options for the cutter.

Vacuum Zoning

ENABLE — Turns on Vacuum Zoning. The system will control the opening and closing of the zone valves.

Not for use on:

- S-91 TCW
- S-91 static or S-91 ZVC cutters without pneumatic zoning
- GTxL

DISABLE — Turns off this feature. When the vacuum generator is on, vacuum is applied to the entire table.

Vacuum Zone Overlap

Sets the distance between the cutter head and the next zone when the system applies vacuum to that zone. (Used only if Vacuum Zoning is enabled.)

Not for use on:

- S-91 TCW
- S-91 static or S-91 ZVC cutters without pneumatic zoning
- GTxL

Vacuum Error Reporting

Turns low vacuum error reporting on and off.

ON — Warnings display if vacuum drops below values set in Vacuum Low Level parameters. Press **Enter** to clear the warning message. If cutting stopped also, press **Start** to resume.

OFF — Turns off this feature.

Vacuum System Parameters

Vacuum Control Type

Choices are:

NONE - Vacuum level is the same regardless of cutter mode.

Not for use on S-91 cutters.

BUTTERFLY - The cutter controls vacuum level based on current cutting mode (idle, cut, or bite).

Not for use on S-91 cutters. Applies only to GERBERcutters equipped with optional Vacuum Level Control system.

VFD BASIC - Vacuum level can be set to High or Low. Not for use on S-91 cutters. Currently, available on GTxL GERBERcutter only.

VFD ENHANCED - The cutter controls vacuum level based on current cutting mode (idle, cut, or bite). Not for use on S-91 cutters. Currently, available on GTxL GERBERcutters equipped with optional Vacuum Level Control system.

EXTERNAL - Vacuum level is the same regardless of cutter mode.

For S-91 cutters only.

Vacuum Level — Bite

Sets the vacuum level that the system maintains when the cutter is biting (conveying the spread for the next bite).

Not for use on S-91 cutters. Applies only to GERBERcutters equipped with a Vacuum Level Control system. Used only if Vacuum Control Type is set to BUTTERFLY or VFD ENHANCED.

Vacuum Level — Cut

Sets the vacuum level that the system maintains when the cutter is cutting.

Not for use on S-91 cutters. Applies only to GERBERcutters equipped with a Vacuum Level Control system. Used only if Vacuum Control Type is set to BUTTERFLY or VFD ENHANCED.

Vacuum System Parameters

Vacuum Level — Idle

Sets the vacuum level that the system maintains when the cutter is idle (the beam, cutter head and conveyor are not moving).

Not for use on S-91 cutters. Applies only to GERBERcutters equipped with a Vacuum Level Control system. Used only if Vacuum Control Type is set to BUTTERFLY or VFD ENHANCED..

X Home Zone1 Offset

Sets the distance from X Home (0,0) to the start of vacuum zone #1.

Not used on:

- S-91 TCW cutter
- S-91 Static or ZVC cutters without pneumatics
- GTxL

Vacuum Low Error Level

Sets the level at which the system displays an error message and halts cutting due to low vacuum.

Vacuum Low Caution Level

Sets the level at which the system displays a warning message about low vacuum level.

Vacuum Control — Pattern Length

Sets the length of time the control signal is applied to the Vacuum Control Valve solenoid. The higher the signal, the longer it is applied.

Vacuum Control — Zero Offset

Sets the duty cycle (Open vs. Closed) of the control signal. The higher the setting, the longer the valve is closed.

Vacuum Control — Vacuum Gain

Sets the response time for the Vacuum Control valve: the higher the setting, the quicker the response.

Number of Vacuum Zones

Sets the number of vacuum zones on the cutting table. Not for use on S-91 or GTxL cutters.

Bite Vacuum Threshold

Sets the vacuum level at which the conveyor begins moving.

Time Until ESM

Sets the number of seconds the cutter is idle before it goes into Energy Savings Mode.

Used only if ESM is enabled. Not for use on S-91 cutters.

Energy Savings Mode

ENABLE — Turns on Energy Savings Mode (ESM).

Not for use on:

- S-91 TCW
- S-91 static or S-91 ZVC cutters without pneumatic zoning

DISABLE — Turns off Energy Savings Mode.

Num of Vac Zones Closed for Bite

Sets the number of table zones closed during a bite, starting from the take-off end of the table.

Not for use on:

- S-91 TCW
- S-91 static or S-91 ZVC cutters without pneumatic zoning
- GTxL

Vacuum Level Valve on Inlet

Identifies whether or not the system has a vacuum level valve: TRUE or FALSE

Not used on S-91 cutters.

Ignore Codes Parameters

Parameters in this category can be used to tell the cutter to ignore or respond to certain marker data codes. The cutter moves the head to the location but does not execute the code.

Ignore Auto Origin (M70) Codes

M70 sets the current X,Y position as the origin.

ON — Cutter ignores M70 codes.
OFF — Cutter responds to M70 codes.

Ignore Aux Drill (M44) Codes

An auxiliary drill is a drill in the auxiliary position on the cutter head.

ON — Cutter ignores auxiliary drill codes.
OFF — Cutter responds to auxiliary drill codes.

Ignore Display Msg (M20) Codes

The M20 code stops cut file processing to display a message on the monitor. For example, an M20 code can be used to display the length and width of the marker before the system starts cutting the marker.

ON — Cutter ignores M20 codes.
OFF — Cutter responds to M20 codes.

Ignore Feedrate (F) Codes

This is used with the POWER Processor option. Makes the system ignore any feed rate codes in the cut data stream.

ON — Cutter ignores F codes.
OFF — Cutter responds to F codes.

Ignore Label Cycle (M31) Codes

ON — Cutter ignores label cycle codes.
OFF — Cutter responds to label cycle codes.

Ignore Codes Parameters

Ignore Lift & Plunge (M46) Codes

The M46 code commands a Lift & Plunge corner at the next data point.

ON — Cutter ignores M46 codes.
OFF — Cutter responds to M46 codes.

Ignore Main Drill (M43) Codes

ON — Cutter ignores standard drill codes. For example, this parameter can be used to continue cutting when the drill is broken. Drills can be cut by hand at a later time.

OFF — Cutter responds to standard drill codes.

Ignore Move Conveyor (M69) Codes

The M69 code tells the system that it has reached the end of the bite.

ON — Cutter ignores M69 codes.
OFF — Cutter responds to M69 codes.

Ignore Overcut Off (M18) Codes

The M18 code turns off automatic Overcut, but keeps Advance Before Plunge.

ON — Cutter ignores M18 codes.
OFF — Cutter responds to M18 codes.

Ignore OV, AP Off (M19) Codes

The M19 code turns off both the automatic Overcut and Advance Before Plunge features.

ON — Cutter ignores M19 codes.
OFF — Cutter responds to M19 codes.

Ignore Pen (D1+ D2) Codes

ON — Cutter ignores pen codes.
OFF — Cutter responds to pen codes.

Ignore Sharpen (M42) Codes

ON — Cutter ignores sharpen codes.
OFF — Cutter responds to sharpen codes.

Bristle Cleaner Parameters

Ignore Slowdown Codes

ON — Cutter ignores slowdown codes including L, M25, and M60-67 codes.

OFF — Cutter responds to slowdown codes.

Bristle Cleaner Parameters

Parameters in this category control:

- Times for cleaning cycle
- Number of sweeps for automatic cleaning

Not for use on S-91 cutters.

Bristl Cleanr (0=N 1=Motr 2=Pneum)

Identifies the type of bristle cleaner available on the cutter:

NONE, MOTORIZED, or PNEUMATIC

Slat Advance Time

Sets the time for a slat on the conveyor to position during automatic bristle cleaning.

Cleaner Head Sweep Time

Sets the time for the bristle cleaner to make one pass on a slat.

Cleaner Head Retract/Plunge Time

Sets the time to plunge or retract the bristle cleaner head.

Bristle Cleaner Sweeps per Slat

Sets the number of times the automatic Bristle cleaner sweeps (passes) each slat on the cutting table.

Number of Table Slats

Sets the number of slats that make up the table cutting surface.

One-Way Bristle Cleaning

ENABLE — Allows the Bristle cleaner to travel in one direction only when sweeping the table.

DISABLE — Turns off this feature.

Knife and Path Intelligence Parameters

Parameters in this category:

- Control how Knife Intelligence compensates for cutting forces on knife.
- Set up Path Intelligence to automatically adjust feed rate for the best cut quality and throughput.

Note: Knife Intelligence parameters are not applicable for these cutters: GTXL, GT 3250, or S-3200.

KI: Gain

Sets the sensitivity of Knife Intelligence. This controls the amount of force available for adjusting the knife heading to correct for lateral forces on the knife. This parameter is not needed for cutting low ply spreads.

Not applicable to GTXL, GT 3250, or S-3200 cutters.

KI: Deflection Angle

Sets the greatest angle Knife Intelligence can turn the C-axis to compensate for lateral force on the knife. This setting depends upon the type and ply height of the material being cut. This parameter is not needed for cutting low ply spreads.

Not applicable to GTXL, GT 3250, or S-3200 cutters.

KI: Max Turn Angle

Sets the maximum knife turn angle the system will execute with KI turned ON. If the angle specified is exceeded, KI will be turned OFF while turning the corner. This parameter is not needed for cutting low ply spreads.

Not applicable to GTXL, GT 3250, or S-3200 cutters.

KI Spring Factor

Set the spring factor used by KI to calculate forces on the knife.

Not applicable to GTXL, GT 3250, or S-3200 cutters.

Knife and Path Intelligence Parameters

KI: Error Detection

Turns Knife Intelligence error detection on or off.

ON — System checks whether the system is receiving a signal from the Knife Intelligence transducer in the presser foot bowl. Not applicable to GTXL, GT 3250, or S-3200 cutters.

OFF — System does not check.

KI: On/Off

ON — Turns on Knife Intelligence when the KI Option is enabled. Not applicable to GTXL, GT 3250, or S-3200 cutters.

OFF — Turns off this feature.

KI: Option

ENABLE — Turns on the KI option. Not applicable to GTXL, GT 3250, or S-3200 cutters.

DISABLE — Turns off this feature.

KI: Fail Time

Sets the interval of time that if no KI activity is sensed, a KI failure error is displayed.

Not applicable to GTXL, GT 3250, or S-3200 cutters.

KI: Corner Disable Time

Sets the amount of time, if any, KI is temporarily shut off while turning a corner with the knife down. Set at 0 (zero) to disable.

Not applicable to GTXL, GT 3250, or S-3200 cutters.

Only Allow Cutting with KI On

ENABLE — Prevents the system from cutting if Knife Intelligence is Off and the Start button is pressed.

DISABLE — Turns off this feature.

Knife and Path Intelligence Parameters

Path Intelligence Option

ENABLE — Turns on the PI option.

DISABLE — Turns off this feature.

PI FR Control — XY Force

ON — Sets the best cutting speed for the material. The system monitors the forces required to move the cutter head and beam during cutting, and adjusts the feed rate accordingly. This feature is controlled by the PI: Optimum Force, PI: Feedrate: Minimum, and PI: Feedrate Maximum parameters.

OFF — Turns off this feature.

PI FR Slowdown — Low Vac

ON — Lowers the feed rate whenever vacuum drops below the level set in the Vacuum Low Caution Level parameter. The system restores the feed rate when vacuum rises by 13 mm/Hg (0.5 in./Hg).

OFF — Turns off this feature.

PI FR Faster — Shp Knife

ON — Temporarily increases the feed rate for a recently sharpened knife above its normal setting. The amount of feed rate increase and the cut distance for which it applies are controlled by field service level parameters.

OFF — Turns off this feature.

PI Optimum Force

Sets the ideal force the system should maintain when the PI: FR Control — XY Force parameter is set for automatic feed rate control. Optimum Force relates to the highest feed rate the system can use without losing cut quality. Optimum Force is affected by feed rate, material, and ply height.

Use the Watch Forces diagnostic to determine the Optimum Force for a given spread.

PI Feedrate Max Increase

Sets the maximum amount by which PI can increase the feed rate at one time.

PI Feedrate Minimum

Sets the minimum feed rate the system uses when the PI: FR Control — XY Force parameter is turned on. This value should be set to where you are currently able to achieve quality cutting. See PI: Feedrate: Maximum.

PI Feedrate Maximum

Sets the maximum feed rate the system uses when the PI: FR Control — XY Force parameter is turned on.

To determine this value, reset in increments above the PI: Feedrate: Minimum value, checking results at each increase to ensure quality.

PI Feedrate Change Ratio

Determines how quickly PI adjusts the cut speed due to XY force. A higher ratio makes the system more responsive to force changes.

PI Velocity to +FR

Sets the velocity rise at which PI will increase the feed rate (cutting speed) for all new contours.

PI Velocity to +FR & Rbld

Sets the velocity rise at which PI will increase feed rate (cutting speed) and rebuild contours.

PI Velocity to -FR

Sets the velocity drop at which PI will decrease the feed rate (cutting speed) for all new contours.

PI Velocity to -FR & Rbld

Sets the velocity drop at which PI will decrease feed rate (cutting speed) and rebuild contours.

PI Force to Stop with Warning

Sets the maximum percentage that the XY Force can be greater than the Optimum Force. If XY Force exceeds this setting, the system stops cutting and warns that there is excessive resistance to head movement.

PI Sharp Knife Full Speedup Dst

Sets the distance the system travels after a sharpening cycle while cutting at the Full Speedup amount.

Knife and Path Intelligence Parameters

PI Sharp Knife Prtl Speedup Dst

Sets the distance the system travels after a sharpening cycle while cutting speed goes from the Full Speedup amount to normal speed.

PI Sharp Knife Speedup Amount

Sets the amount PI increases normal cutting speed to operate at the Full Speedup amount after a sharpening cycle.

PI Vacuum Low Slowdown Factor

Sets the amount PI reduces normal cutting speed after the vacuum level falls below the Vacuum Low Caution Level.

Knife Break Force

Sets the force (required to move the knife through the spread) at which the system generates a warning about knife breakage. The Knife Break Risk Point parameter is also considered.

Knife Break Risk Point

Sets the percentage of knife sharpenings before the system generates a warning about knife breakage. The Knife Break Force parameter is also considered.

Data Format/Scaling and RSO

Data Format/Scaling and RSO

Parameters in this category:

- Define the format of cut file data or AAMA data conversion.
- Define X and Y-axis scaling.
- Define marker size.
- Initialize RSO cutters.
- Enter correction value for Center Line parameters.

X Data Scale Factor

Changes the X-axis scaling to adjust the size of the marker to the size of the available material. The range of change permitted is 0.800 to 1.200 (80 to 120 percent) of the existing coordinate values in the marker.

To determine the X Data Scale Factor, divide the length of usable material in the X-direction (spread length) by the marker length.

For example, if the marker is 5.00 m long and the material is only 4.95 m long:

1. Divide the usable material length by the marker length: $4.95 / 5.00 = 0.9900$.
2. Round down the Data Scale Factor to three decimal places. In this example, set X Data Scale Factor to 0.990.

CAUTION: The X Data Scale Factor does not reset automatically. Use either the Settings or Parameter Manager command to reset the X Data Scale Factor to 100 percent. If you do not reset the Data Scale Factor, you may improperly cut other markers.

Y Data Scale Factor

Changes the Y-axis scaling to adjust the size of the marker to the size of the available material. The range of change permitted is 0.800 to 1.200 (80 to 120 percent) of the existing coordinate values in the marker.

To determine the Y Data Scale Factor, divide the width of usable material in the Y-direction (spread width) by the marker width.

For example, if the marker is 5.00 m wide and the material is only 4.95 m wide:

1. Divide the usable material width by the marker width: $4.95 / 5.00 = 0.9900$.
2. Round down the Data Scale factor to three decimal places. In this example, set Y Data Scale Factor to 0.990.

CAUTION: The Y Data Scale Factor does not reset automatically. Not resetting the Data Scale Factor, may improperly cut other markers. Use either the Settings or Parameter Manager command to reset the Y Data Scale Factor to 100 percent.

Data Format

Identifies the format of the cut file data to the system. The format you select must match the format used in your cut files. There are three formats:

- 3.3 ENGLISH
- 4.2 ENGLISH
- 5.1 METRIC

The digit before the decimal point shows how many places are reserved for the whole number. The digit after the decimal point shows how many places are reserved for the decimal.

Note: This parameter does not select the units of measure (metric or English) displayed at the operator console. Units of measure are set by the Metric Display parameter.

Data Format/Scaling and RSO

Marker Size: Use Auto Calculate

ON — The system uses the automatically calculated length and width of the marker whenever needed, for example, Check Straightness, Check Origin, or Right Side Option.

OFF — The system prompts for new marker size.

Note: If you set both this parameter and the Marker Size: Use M20 Code parameter to ON, the system uses the M20 code.

Hints on Using with Centerline Origin for Automatic Tube Centering

M20 code can contain the width of the spread for which the marker was designed. If this parameter is turned off, C-200MT uses the width of the marker to determine centerline origin. Since many markers are either wider or narrower than the spread from which they are cut, an incorrect origin could result unless this parameter is turned ON when using the Automatic Tube Centering system.

Marker Size: Use M20 Code

ON — The system uses the marker length and width in the M20 code whenever needed, for example, Check Straightness, Check Origin, or Right Side Option. The software supports M20 codes from AM-5, AccuMark, MicroMark, and Lectra CAD systems.

OFF — Turns off this feature.

CAUTION: Do not set this parameter to ON if your markers do not use M20 codes. The system will not calculate the origin properly.

Note: If you set both this parameter and the Marker Size: Use Auto Calculate parameter to ON, the system uses the M20 code.

Right Side Initialize

Initializes a cutter that has the Right Side Option (RSO). After initializing on the operator's side of the cutter, the head moves to the position where:

$X = 5$

$Y = \text{usable table width} - 5$

AAMA Data Conversion

ENABLE -Converts AAMA (American Apparel Manufacturer's Association) data automatically at the cutter. Use only when cutting AAMA data.

DISABLE — Turns off this feature.

Center Line Calibration

Sets a value to calibrate the system's calculation of the origin when using Center Line Origin parameters.

Bite Parameters

Bite Parameters

Parameters in this category:

- Define how the cutter handles bites.
- Define end park position.
- Are not for use on S-91 Static systems.

Allow Piece Reorder

Determines whether Run Time Biting reorders piece cutting sequence in order to maximize bite lengths when Variable Length Biting is ON (default).

ON — Preprocesses all drills for pieces on the cutting table before cutting any of the pieces.

OFF- Turns off this feature.

Run Time Biting

ON — Run Time Biting processes markers, turning them into bite markers for cutting on the GERBERcutter. Use to:

- Bite and cut markers that do not have bite information
- Change bite information inserted by the marker maker

OFF — Turns off this feature. Remember to do this when cutting markers that have bites added using:

- SBITE utility in DOS
- Bite Generation in the PreProcessor utility

Not for use on S-91 Static systems.

Variable Length Biting

ON — Turns on Variable Length Run Time Biting. Pieces will only be split if they exceed the table surface.

OFF — Turns off this feature. Static Length biting is used instead.

Minimum Bite Length

Sets the length of a conveyor advance during a Static Length bite.

Maximum Bite Length

Sets the maximum length of the bite window during a Static Length bite or the maximum length of a conveyor advance for a Variable Length bite.

Pause Before Convey

ON — Stops before conveying each bite off the table. Press **Start** to continue.

OFF — Turns off this feature.

Pause Between Bites

ON — Stops at the end of each bite to:

- Reposition the spread
- Use the Check Straightness command to reset the origin
- Use the Fabric Alignment command to set a new Fabric Alignment angle

OFF — Turns off this feature.

Convey Last Bite Off Table

ON — When Contiguous Cut is ON, the system moves the last bite off the cutting table after it is finished processing the marker.

OFF — Turns off this feature.

Bite Parameters

Bite Clip Start Position

Sets the start position (Y coordinate) of the bite clip. A bite clip is:

- A cut in the fabric that marks the end of a bite
- Outside the marker
- Perpendicular to the X-axis

This parameter can be used to manually relocate a bite in case of bite error during cutting.

Note: A negative value is acceptable.

Bite Clip End Position

Sets the end position (Y coordinate) of the bite clip.

Bite Park Offset

Sets the distance from the edge of the table where the cutter head moves to while the conveyor is advancing a bite. This parameter is related to Bite Park Location.

Insert M23 Edge Clip Code

ENABLE — Identifies edge clips during Run Time Biting.

DISABLE — Turns off this feature.

Bite Veeroff

Sets the length of veeroff in Y when overlapping for split pieces. This ensures the split piece is fully cut out. This parameter is related to Bite Overlap.

Marker End Park (0=None 1= -Y 2= +Y)

Determines the location where the cutting head parks after completing a marker.

NONE — Wait at current point

-Y — Move to extreme -Y location

+Y — Move to extreme +Y location

This parameter is related to Marker End Park Position X and Marker End Park Position Y.

Marker End Park Position X

Determines the +X or -X position where the cutting head parks after completing a marker. This parameter is related to Marker End Park (0=None, 1= -Y, 2= +Y).

Marker End Park Position Y

Determines the +Y or -Y position where the cutting head parks after completing a marker. This parameter is related to Marker End Park (0=None, 1= -Y, 2= +Y).

Maximum Bite Adjustment

Sets the amount of adjustment the system can correct for after a bite advance without generating an error.

Bite Park Location

Controls which side of the cutting table a bite park occurs. Bite park is the place that the cutter head moves to when the conveyor advances a bite.

ON — The cutter head moves to the minus (left) side of the cutting table.

OFF — The cutter head moves to the plus (right) side of the cutting table.

Note: This parameter is related to Bite Park Offset.

Bite Parameters

Cut Small Pieces First

Cut files are normally sequenced by CAD systems to cut the pieces in a sinusoidal curve or lazy M. However, most marking systems can order a change in sequence to provide for cutting of small pieces before the larger pieces are cut. Normally, it is the area of the piece that determines its category. When Run Time Biting recalculates the marker to insert bites, it can change the order in which pieces will be cut. For that reason, there are two parameters that can be set at the cutter to maximize cut quality of small pieces during Run Time Biting:

- Cut Small Pieces First
- Max Area of Small Piece

Utilizing these parameters properly during run time biting can improve small piece cut quality.

The Cut Small Pieces First command instructs the system to first cut all small pieces in a cut file. The system must also know the Maximum Area of the Small Piece the system will consider small.

Max Area of Small Piece

Marker makers can order a change in sequence to provide for cutting of small pieces before the larger pieces are cut. Normally, it is the area of the piece that determines its category. When Run Time Biting recalculates the marker to insert bites, it can change the order in which pieces will be cut. For that reason, there are two parameters that can be set at the cutter to maximize cut quality of small pieces during Run Time Biting:

- Cut Small Pieces First
- Max Area of Small Piece

Utilizing these parameters properly during run time biting can improve small piece cut quality.

The Max Area of Small Piece parameter reflects the maximum piece area the system will consider small for a Cut Small Pieces First command.

Recalculate Small Pieces

ON – Cuts small pieces at the beginning of a bite until the first large piece is cut.

OFF – Turns off this feature.

Bite Overlap

If a piece lies across a bite boundary, the piece is cut in two steps. At the end of the first bite, the knife veers off the cut line. At the beginning of the second bite, the knife veers into the cut line.

Use with Bite Veeroff. The X distance of the veeroff equals half of Bite Overlap; the Y distance equals Bite Veeroff.

Conveyor Table Settle Time

Sets the amount of time the system will wait for the conveyor to completely stop moving at the end of a conveyor advance before the origin is adjusted and cutting continues.

Continuous Cutting

This parameter turns on the pay-for Continuous Cutting option. The Continuous Cutting Installed parameter must be set to YES.

ON — The cutter continues cutting, beginning the next bite while the conveyor is moving the previous bite off the table.

OFF — The cutter pauses when it finishes processing a bite.

Not for use with S-91 cutters.

Continuous Cutting Installed

This Field Service level parameter notifies the C-200MT program whether or not the Continuous Cutting option is installed. In order to use Continuous Cutting, the system must:

- Be equipped with a bite encoder mounted under-the-table (not on the beam).
- Have a variable speed (DC) CTOT motor.

Note: Bite lengths must be set to no more than one half of the default settings for the system to prevent off-table errors.

YES — The C-200MT program supports continuous cutting while the conveyor is advancing **as a pay-for option**. The effect Continuous Cutting has on throughput depends on piece size. If the marker contains small pieces, this feature can have a positive impact on overall system throughput. However, because the usable cutting surface is effectively halved, continuous cutting is not recommended for markers that contain large pieces (longer than one half of the usable table surface).

Parameter settings required for Continuous Cutting:

- Continuous Cutting Installed must be set to YES.
- Continuous Cutting must be set to ON.
- Conveyor Encoder Location must be set to TABLE.
- Variable Speed CTOT must be set to ON.

NO — This option is not available.

Options and Modes

Options and Modes

Parameters in this category turn on/off various GERBERcutter features.

Use Edge Finders

YES — When Centerline Origin Edge Points parameter is turned ON, Automatic Tube Centering is used to find the origin and fabric alignment for a tubular spread.

NO — This feature is turned OFF.

Edge Finder Auto-Alignment

ON — The cutter automatically checks origin and fabric alignment for all bites in the cut file.

OFF — This feature is turned OFF. The cutter checks origin and fabric alignment for the first bite only and waits for the operator to manually check origin and fabric alignment for each successive bite.

Log Edge Finder Info

ON — A log of events for the Automatic Tube Centering system is recorded in the edge.log file in the c:\report directory.

OFF — This feature is turned OFF.

Auto-Scale Y

ON — Automatically scales data in the + or — Y direction while using the Automatic Tube Centering system.

OFF — This feature is turned OFF. The Centerline Origin Edge Points parameter must be turned ON.

Max Auto-Scale Y Change

When the Auto Scale Y parameter is turned ON, enter a plus or minus percentage of change allowable for automatic data scaling. The Centerline Origin Edge Points parameter must be turned ON.

Dry Run Tool is Light

ON — When Dry Run mode is selected, the light is positioned at the location where the knife would be, if cutting, and traces the cut path. This may be easier to see than if the knife is used as the tool.

OFF — When Dry Run mode is selected, the raised knife traces the cut path.

Clear Origin on Cutfile Close

ON — When a cut file is closed the origin returns to X0, Y0.

OFF — Turns off this feature.

Installed Edge Finders

NONE — Neither of the two edge finders is installed for the Automatic Tube Centering system.

-Y — The edge finder closest to the operator console is installed.

+Y — The edge finder farthest from the operator console is installed.

BOTH — Two edge finders for the Automatic Tube Centering system are installed.

Saving of Parameter Changes

This service level parameter determines whether or not parameter changes made at the user level can be permanently saved to setup files.

ENABLE — Prevents parameter changes from being saved to setup files.

DISABLE — Turns off this feature.

Options and Modes

Single Step

ON — Cutter executes one block of cut data at a time. This ability is helpful when recovering from an error or trying to determine what caused an error during cut file (marker) processing.

CAUTION: Do not use this mode to cut a marker because the C axis is not working.

Note: During Single Step mode, the cutter head rises to prevent fusing if more than three seconds elapse without initiating a command. The cutter head lowers when you initiate the next command.

OFF — Cutter executes data normally.

Marker Flip

ON — Rotates a single (or the lower marker for marker stacking) 180 degrees during processing. This feature could be used, for example, to cut corduroy that was laid in the wrong direction.

Because Marker Flip reverses the order in which the pieces are cut, you can use Marker Flip on a conveyORIZED system. The system cuts the marker from the take-off end to the take-on end as usual.

OFF — Turns off this feature.

Note: You must turn on Run Time Biting to use Marker Flip. Also, Marker Flip must be turned ON before selecting a marker.

Note: Unlike Marker Flip, Mirror Image does not change the order in which the pieces are cut. Because Mirror Image cuts the marker from take-on end to take-off end, you can not use Mirror Image on a conveyORIZED system.

Marker Flip Top

ON — Rotates the top marker (for marker stacking) 180 degrees during processing; reorders pieces the same as Marker Flip. This feature could be used, for example, to cut corduroy that was laid in the wrong direction.

OFF — Turns off this feature.

Marker Stacking Reorder

YES — Reorders the piece numbers in both stacked markers so they are cut as if in one marker.

NO — Cuts the entire bite of bottom marker and then the entire bite of top marker.

Marker Stacking Reorder Channel

Sets the width of the channel in which pieces must fit when reordering for stacked markers.

Language Support

1 — Enables Native language support by telling the system to use translated messages from the C:\SYSFILES\NATIVE directory.

0 — Turns off this feature.

Kanji Character Set

ENABLE — Changes display settings for the operator console monitor in order to display Kanji characters in the C-200MT program.

DISABLE — Turns off this feature.

Check Origin

ON — Ensures that the marker fits completely on the usable cutting surface.

OFF — Turns off this feature. Check Origin must be off in order to use Fabric Alignment.

If a marker positioned incorrectly or the origin setting is inappropriate, Check Origin gives you a range of acceptable range of X and Y values for the origin. This happens right after you press Start. A message displays similar to the following:

Bad Origin: Y too high (50.00)
acceptable range: 30.000 — 45.055

In this example, set a new origin with a Y value that falls between 30.000 and 45.055. The marker will now fit within the usable cutting surface.

Note: The message displayed depends on the size of the marker, the origin, and the data format.

Options and Modes

Verify Keyboard Entry

ON — Displays a dialog box during certain procedures to verify keyboard entry.

OFF — Turns off this feature.

Block Delete

ON — The cutter does not cut lines that have Block Delete codes inserted from the GPP POWER Processor.

OFF -The cutter ignores Block Delete code.

Cut/Plot Mode

Determines the tool to use during automatic processing.

CUT — Selects the knife.

PLOT — Selects the pen.

Dry Run

ON — Allows automatic cutting, including C-axis rotation, with the knife up. Use when recovering to verify that the correct marker is selected.

OFF — Moves the cutter head with the knife down to cut the marker.

Note: You can not choose Dry Run while in Plot mode.

Metric Display

Determines how measurements display in the C-200MT program:

OFF — Display English measurements

ON — Display Metric measurements

Delete Cut File After Cutting

ENABLE -Deletes cut files (markers) from the system drive after cutting.

DISABLE — Turns off this feature.

CAUTION: If cutting from a network, network files will be deleted. Also, to avoid losing data, make sure cut files are saved on disk or main frame when this parameter is turned on.

Re-Sealer Option

ENABLE — Tells the system to use the optional Resealer. Available for S-5200 and S-7200 cutters.

DISABLE — Turns off this feature.

Resealer Up For Continuous Cut

ON — Standard Releaser logic is turned off; Resealer remains continually in “up” position to cut lofty goods. Available for S-5200 and S-7200 cutters.

OFF — Standard Resealer logic is active.

Preprocess Drills

ON — Tells the system to process all drills in the bite before cutting pieces. Using this option saves wear and tear on the optional electric drill.

OFF — Turns off this feature.

Center Line Origin Center Point

ON — Centers the cut of tubular goods relative to a center stripe line or other fabric pattern. When using Center Point, move the origin light to the center of the spread. Then set the origin. The system determines center line origin by subtracting half of marker width from the origin just set.

OFF — Turns off this feature.

Options and Modes

Center Line Origin Edge Points

ON — Centers the cut of tubular goods by calculating a center line origin based on edge points. When using Edge Points, move the origin light to the bottom corner of the spread and then set the origin. Repeat for the top corner of the spread. The system determines center line origin by subtracting half of marker width from a system- selected point half way between the upper and lower points.

OFF — Turns off this feature.

Transfer Markers When Selected

Transfers markers from a network to the /TEMP directory in order to ensure uninterrupted cut data transmission.

OFF — Markers are not transferred.

COPY — Markers are copied to the cutter's hard drive.

MOVE — Markers are moved to the cutter's hard drive.

Y Axis Edge Align (0=N 1=+Y 2=-Y)

Aligns cutter head with edge of spread.

0 — Not installed

1 — +Y side

2 — -Y side

Bar Code Reader Option

ON — Tells the system to use the optional Bar Code Scanner to read cut file and setup information.

OFF — Turns off this feature.

Cutter Info Database Option

ENABLE -Turns on this optional program to automatically save information to the CIDB database while a marker is being cut.

To use this option

1. Enable this parameter and save the setup file.
2. Quit the C-200MT program.
3. Restart the C-200MT program.

DISABLE — Turns off this feature.

Mirror Image and Clipping Window

Mirror Image and Clipping Window

Parameters in this category:

- Turn on the mirroring options which rotate the marker before cutting.
- Turn on the clipping window feature and define boundaries.

Mirroring Option

ENABLE — Allows you to use Mirror Image mode or Y-axis Mirror Image mode to rotate a marker during processing. Pieces are not reordered.

DISABLE — Turns off this feature.

Note: If you disable Mirroring Option, the other mirroring parameters are not valid.

Note: To reorder pieces when rotating a marker 180 degrees, use the Marker Flip parameters instead.

Mirror Image

ON — Rotates a marker and cuts a mirror image of the marker; pieces are not reordered. This parameter might be used in the aerospace industry to cut a wing. Mirroring Option must be enabled in order to use this parameter. For use on S-91 Static cutters only.

When Y-Axis Mirror Image is OFF, this feature rotates a marker 180 degrees and cuts pieces from the take-on end of the table to the take-off end. For example, if piece number 1 is normally cut at the lower left (near to the operator), it would still be cut first but in the upper right corner of the table. Use only for markers that fit entirely on the cutting surface.

When Y-Axis Mirror Image is ON, this feature rotates the marker along the Y-axis only; pieces are not reordered. For example, if piece number 1 is normally cut at the lower left (near to the operator), it would still be cut first but in the upper left corner of the table. Can be used for markers that are longer than the cutting surface.

OFF — The system cuts the marker normally (without mirroring).

Y-Axis Mirror Image

Rotates the marker along the Y-axis only; it is useful for processing different markers that have the same width. The marker can be longer than the usable surface.

ON — Uses Y-axis Mirror Image mode for the marker. Mirroring Option must be enabled and Mirror Image must be turned ON; also define Y-Axis Mirror Reflection Line.

OFF — Uses Mirror Image mode for the marker.

Y-Axis Mirror Reflection Line

Sets the reflection line for Y-Axis Mirroring. Set this equal to the width of the marker. Used when Y-Axis Mirror Image is ON.

Note: Every marker using this setup file must have the same width.

Clipping Window Option

ON — Eliminates unwanted travel with the knife down outside the folded edge of goods. Lets you set a low and a high boundary so the system will not cut fabric outside these boundaries.

OFF — Turns off this feature.

Clipping Low Boundary

Sets a low clipping boundary in the -Y direction from the origin. Clipping Window Option must be ON.

Clipping High Boundary

Sets a high clipping boundary in the +Y direction from the origin. Clipping Window Option must be ON.

Clipping Low Boundary (Top Mrk)

Sets a low clipping boundary in the -Y direction from the origin for the top marker when using Marker Stacking. Clipping Window Option must be ON.

Clipping High Boundary (Top Mrk)

Sets a high clipping boundary in the +Y direction from the origin for the top marker when using Marker Stacking. Clipping Window Option must be ON.

OPSTOP and Report Options

Parameters in this category:

- Control how cutter responds to Opstop codes.
- Define Cutter Info Database reporting.

Camera Extension Distance

Not currently in use by the AccuMatch system.

Use AccuMatch

ENABLE — For any cut file that contains Opstops, the AccuMatch system is used to capture images and match to the fabric.

DISABLE — This feature is turned OFF. The GERBERcutter pauses whenever an Opstop is encountered in a cut file and waits for the operator to manually check and adjust match points.

For use on GT 3250, S-3200, or GTxL systems only.

OPSTOP Section Processing

The Run Time Biting (SBITE) software can perform special processing of "sectional" Opstops placed in "empty" pieces by Microdynamics OPTICUT systems. When this parameter is enabled, SBITE does the following:

- As soon as one section is fully cut, SBITE conveys the next section as far left as possible before beginning to process it. This minimizes the chances of the section's opstop point not yet being on the table.
- No matter which piece in the current section contains the opstop, SBITE checks that the opstop is processed before any pieces in that section are cut.

Note: There should be exactly one opstop per section.

ON — This type of Opstop is processed before any pieces in the section are cut. These Opstops will be moved into the piece (in the section) whose leading (-X) end point is closest to the take-off end of the table.

OFF — Turns off this feature.

OPSTOP and Report Options

Not for use on S-91 Static systems.

OPSTOP

ENABLE — The cutter responds to M01 (Optional Stop) codes in the marker. When the cutter reads a M01 code, it stops the cutting to allow plaid/stripe matching. Slew the beam to align and press Start to resume cutting.

DISABLE — The cutter ignores M01 codes.

Preset OPSTOP

ON — Before cutting begins, the cutter head moves to each Opstop point defined in the marker so you can make any necessary adjustments for plaid/stripe matching. This optimizes throughput during actual cutting.

OFF — Turns off this feature.

Regional OPSTOP

ENABLE — Adjustments made on a piece at an Opstop apply to all successive pieces until the system encounters the next Opstop.

DISABLE — Turns off this feature.

OPSTOP Slow Slew

Provides additional time for making long adjustments when using Opstops.

ON — The cutter automatically enters slow slew operation (feed rate 1) when stopped for an OPSTOP.

OFF- The slew speed for an OPSTOP is determined by the feed rate setting.

Cutter Info Database Output

FILE — Writes report data in the Cutter Info Database to the C:\REPORT\MARKER*.RPT file.

OFF — Turns off this feature.

Selvage Cut/Contiguous Cut

Selvage Cut/Contiguous Cut

Parameters in this category:

- Control Selvage/Web cutting for vertical line cuts through the selvage or the selvage and web.
- Control processing of consecutive markers.

Selv/Web (0=Off 1= Selv 2 = Web&Sel)

Automatically cuts a vertical line through the selvage on both sides of the material, or through both the selvage and the web (uncut material) between pieces. The selvage/web cut occurs after the bite edge clip, but before the conveyor advance.

Performance is controlled by the Selvage Cut Position, Selvage Cut Size – Bottom, Selvage Cut Size – Top, and Minimum Web Cut Size parameters.

OFF — Turns off selvage/web cutting.

SELVAGE — Turns on selvage cutting only.

WEB&SELV. — Turns on both selvage and web cutting.

Selvage Cut Position

Determines -X distance from the bite clip to where the selvage/web cut occurs.

Selvage Cut Size – Bottom

Determines how far the selvage cut extends below the actual marker.

Selvage Cut Size – Top

Determines how far the selvage cut extends above the actual marker.

Minimum Web Cut Size

Sets the minimum thickness allowable for a web cut to be made. If the distance between any pieces on the selvage/web cut line is less than this setting, no web cut occurs.

Intermarker Gap

Sets the distance between the end of one marker and the beginning of the next. For example, can be used when Continuous Cut is ON.

Minimum Size of First Bite

When Contiguous Cut is ON, sets the minimum table length that must be available for the next marker to be processed without conveying cut goods off the cutting table. This parameter works only when Convey Last Bite Off Table is OFF.

Contiguous Cut

ON — The system automatically sets an origin for the next marker a predetermined distance prior to the end of the current file. This allows the cutter to continually cut markers when using a job queue.

OFF — The system stops when it finishes processing a marker.

System Setup Parameters

Parameters in this category set maximum beam velocities for cutting and are used by Field Service to setup systems.

Beam Square Adjustment

System generated value based on the distance between index positions for x1 and x2 as determined by the Beam Square Test and Beam Squareness diagnostics.

Beam Squared

YES indicates that the Beam Square Adjustment value is valid; NO indicates the beam is ready to be squared. This parameter can be set to NO using the Beam Squareness diagnostic.

Bite Error Adjustment

Sets distance between bites, independent of the automatic bite adjustment. Used only on cutters with Dual Beam option.

Maximum Cut Velocity

Sets the maximum beam velocity when the beam moves in a straight line with the knife down. This value is in inches or centimeters per minute.

Max Cut Acceleration

Sets the maximum acceleration for X and Y axes allowed during cutting. This parameter can be set to reduce vibration of the knife.

C Axis Sharpen Speed

Sets C-axis rotation velocity during sharpening. Not for use on GT 3250 OR S-3200 cutters.

X-Axis Stepsize

Sets the size of one step as read by the X-axis encoder; can be seen as a square wave on an oscilloscope.

System Setup Parameters

Y-Axis Stepsize

Sets the size of one step as read by the Y-axis encoder; can be seen as a square wave on an oscilloscope.

Usable Table Distance in X

Sets the usable distance on the table in the X direction (from position 0,0 to +X).

Usable Table Distance in Y

Sets the usable distance on the table in the Y direction (from position 0,0 to +Y).

C Home to System 0 Degrees

Sets the number of degrees between the C-axis home position and 0 degrees. Reset this parameter whenever the head is changed.

Offset in X — Knife

Sets the offset distance of the knife from the home position in the X direction (from origin to the center of the tool).

Offset in Y — Knife

Sets the offset distance of the knife from the home position in the Y direction (from origin to the center of the tool).

Offset in X — Drill 1

Sets the offset distance of the first drill from the home position in the X direction (from origin to the center of the tool).

Offset in Y — Drill 1

Sets the offset distance of the first drill from the home position in the Y direction (from origin to the center of the tool).

Offset in X — Pen

Sets the offset distance of the knife from the home position in the X direction (from origin to the center of the tool).

System Setup Parameters

Offset in Y — Pen

Sets the offset distance of the knife from the home position in the Y direction (from origin to the center of the tool).

Ramp Down Dist — Auto Bite

Sets the distance from the end of the bite where the DC conveyor motor begins to step down voltage until the conveyor is stopped during an Automatic bite.

Not for use on S-91 cutters.

Offset in X — Drill 2

Sets the offset distance of the auxiliary drill from the home position in the X direction.

Offset in Y — Drill 2

Sets the offset distance of the auxiliary drill from the home position in the Y direction.

Offset X — Camera

Sets the offset distance of the camera for the AccuMatch system from the home position in the X direction.

Offset Y — Camera

Sets the offset distance of the camera for the AccuMatch system from the home position in the Y direction.

Offset Edge Sensors X

Sets the offset distance of the edge sensor for the Automatic Tube Centering system from the home position in the X direction.

Offset Bottom Edge Sensor Y

Sets the offset distance of the edge sensor nearest the operator console for the Automatic Tube Centering system from the home position.

Offset Top Edge Sensor Y

Sets the offset distance of the edge sensor farthest from the operator console for the Automatic Tube Centering system from the home position.

Max Extension -Y Edge Finder

Sets the maximum number of encoder counts that the -Y edge finder can move toward the +Y edge finder. Use with the Automatic Tube Centering option.

Max Extension +Y Edge Finder

Sets the maximum number of encoder counts that the +Y edge finder can move toward the -Y edge finder. Use with the Automatic Tube Centering option.

Cut File Preprocessor Parameters

Parameters in this category are used with the PreProcessor command in the Utilities menu.

Length of Reverse Segment

Sets the distance from the folded edge to plunge the knife before cutting the fold.

Edge Clipping Angle

Sets the minimum angle between the folded edge and the segment of the piece at that fold.

Knife Width

Sets the size of the knife, measured from the cutting edge to the heel.

Gap to Ignore

Sets the smallest gap that will be ignored during cut path optimization.

Distance to Corner

Sets the length from the nearest corner of a piece to the point where the knife enters the piece to prevent cutting over a corner. This value should not be less than the knife width.

Path Overlap

Sets the amount of overlap at the transfer points two pieces. This amount will automatically be inserted before and after the transfer points of both pieces.

Corner Angle

Sets the maximum inside angle to consider the point as a corner. An inside angle is 180 degrees minus the change in knife heading.

Cut File Preprocessor Parameters

Transfer Angle

Sets the minimum angle for the transfer point. When optimizing cut path, the system makes sure that the angle of entry into another piece is larger than this value.

Max Length to Ignore

Sets the largest line segment that can be ignored after removing common lines.

Max Distance with Knife Down

Sets the maximum distance between two points with the knife down; this parameter should match the Maximum Segment Length parameter.

Message Stop

YES — Adds M20 codes to a cut file.

NO — Turns off this feature.

Max # Points Per Piece

Sets the maximum number of points allowed per piece.

Overlap Distance

Not used in this software release.

Smallest Piece

Sets the maximum area for a piece which the system will consider as being small.

Percent of Piece

Sets the % of a piece that must be included in a window in order to be cut with other pieces in that window.

Queue Manager Parameters

Queue Manager Parameters

Parameters in this category relate to the Queue Manager utility.

Auto Start Jobs in Queue

ON — The system automatically processes all jobs in a Queue Manager job queue without stopping.

OFF — Turns off this feature. The system stops between jobs in the queue.

Del Job From Queue After Cutting

YES — Each job entry is deleted from the job queue file immediately after being processed; the cut file is not deleted.

NO — Turns off this feature.

Del Queue After Cutting

Determines whether a job queue and/or cut files in it are deleted immediately when done cutting the queue.

QUE FILE — Queue file is deleted only.

MARKERS — Markers in a queue are deleted but not the queue file.

QUE&MARKER — The queue file and all markers in the queue are deleted.

OFF — Turns off this feature.

Read Queue Name Bar Codes

YES — When this feature is turned on, all bar codes read (except setup files *.CNF) are assumed to be job queues and are automatically opened for running.

NO — Bar codes read will be opened as single cut files.

Security Parameters

Parameters in this category control the ability to change or edit Cutter and Notch parameter tables when opening an AccuMark marker for cut processing.

User Parameter Table - Selecting

When this Security parameter is ENABLED, an additional dialog box displays after choosing the AccuMark marker you want to open. This dialog box allows you to select different Cutter and Notch parameter tables for the current marker.

Note: To make edits to these parameter tables, the User Parameter Table - Editing parameter must also be enabled.

User Parameter Table -Editing

When this Security parameter is ENABLED, an additional dialog box displays after choosing the AccuMark marker you want to open. This dialog box allows you to edit the original Cutter and Notch parameter tables for the current marker.

Note: To select different parameter tables, the User Parameter Table - Selecting parameter must also be enabled.

Field Service Level Categories and Parameters

Field Service Level Categories and Parameters

Some categories and parameters in the following sections can only be accessed using the Field Service password.

Note: Additional Field Service level parameters are included in categories that can be accessed at any level. These parameters are documented with the applicable category.

Field Service Level categories and parameters in these categories are listed in the table below.

Cutter Type

Dual Beam

ENABLE — Turns on the Dual Beam option. For S-91 ZVC cutters only.

DISABLE — Turns off this feature.

AccuMatch Installed

Identifies whether or not the system supports the AccuMatch system for plaid/stripe matching:

NO or YES

Table Length (0=Standard,1=Long)

Identifies the length of the cutting surface for GTxL, GT 3250/5250/7250 and S-3200/5200/7200 cutters:

STANDARD or LONG

Beam Width

Identifies the width of the beam:

1.7m, 2.0m, 2.4m (for GTxL, GT 3250, OR S-3200/5200/7200 cutters)

3.0m (for S-91 cutters except 10 ft. S-91 Static)

3.3m (for GT 5250/7250 and S-5200/7200 cutters)

Right Side Control

ENABLE — Turns on the Right Side Option (RSO).

DISABLE — Turns off this feature.

Field Service Level Categories and Parameters

PC Serial Communication

MCC Communications Port

Selects the communications port for PC to MCC communications; choices:

COM1, COM2, COM3, or COM4.

MCC Comm Port Beam 2

Selects the communications port for PC to MCC communications for the second beam on an S-91 cutter with Dual Beam option; choices:

COM1, COM2, COM3, or COM4.

Bar Code Reader Com Port

Sets the communications port for the optional Bar Code Reader; choices:

COM1, COM2, COM3, or COM4.

Dry Haul Speeds

Projector Output Port

Sets the communications port for the seamless interface between the GERBERcutter and third-party projector software.

Maximum Dry Haul Velocity

Sets the maximum velocity during automatic cutting when the tools are up.

Dry Haul Acceleration

Sets the dry haul acceleration during automatic cutting when the tools are up.

Field Service Level Categories and Parameters

C-Axis

C-Axis delay

Sets the amount of delay at the end of a C-axis move before the system starts the next move.

C-Axis cut velocity

Sets the maximum velocity of C-axis rotation during cutting.

C-Axis max velocity

Sets the maximum velocity of the C-axis when turning corners.

C-Axis Acceleration

Sets the acceleration of the C-axis in radians per second squared.

C Axis Settle Time

Determines the amount of time in seconds the system waits for the C-axis to settle at the completion of a C-axis turn. This parameter may have to be increased to prevent C-axis following errors when the number of sharpen cycles is set higher than 8 on an S-91 system.

XY Axes

X Home to System 0

Sets the X-axis 0.00 inch position a specific distance from the -X home switch.

Y Home to System 0

Sets the Y-axis 0.00 inch position a specific distance from the -Y home switch.

XY-Axes delay

Sets the amount of wait time after an XY move is made.

Cutfile Percentage Primary Beam

Sets the percentage of the spread currently on the table which is cut by the Primary beam on a cutter with the Dual Beam option.

Field Service Level Categories and Parameters

Limit Switches

Lm Switch Status — Bite Sensor

Sets the status of the bite sensor limit switch. Only reports status of limits; mechanical stops actually limit.

NONE — No limit switch.

UP — Just the Up limit switch.

Not used on S-91 cutters.

Lm Switch Status — Drill 1

Sets the status of the drill limit switch. Only reports status of limits; mechanical stops actually limit.

NONE — No limit switch.

UP — Just the Up limit switch.

DOWN — Just the Down limit switch.

BOTH — The Up and Down limit switches.

Lm Switch Status — Knife

Sets the status of the knife limit switch. Only reports status of limits; mechanical stops actually limit.

NONE — No limit switch.

UP — Just the Up limit switch.

DOWN — Just the Down limit switch.

BOTH — The Up and Down limit switches.

Lm Switch Status — Pen

Sets the status of the pen limit switch. Only reports status of limits; mechanical stops actually limit.

NONE — No limit switch.

UP — Just the Up limit switch.

DOWN — Just the Down limit switch.

BOTH — The Up and Down limit switches.

Lm Switch Status — Sharpen Arm

Sets the status of the sharpen arm limit switch. Only reports status of limits; mechanical stops actually limit. Used on GT 3250 OR S-3200 cutters only.

NONE — No limit switch.

UP — Just the Up limit switch.

DOWN — Just the Down limit switch.

BOTH — The Up and Down limit switches.

Lm Switch Status — Drill 2

Sets the status of the auxiliary drill limit switch. Only reports status of limits; mechanical stops actually limit.

NONE — No limit switch.

UP — Just the Up limit switch.

DOWN — Just the Down limit switch.

BOTH — The Up and Down limit switches.

Limit Switch Status — Edge Sensor

Sets the status of the edge sensor limit switch. Only reports status of limits; mechanical stops actually limit.

NONE — No limit switch.

UP — Just the Up limit switch.

DOWN — Just the Down limit switch.

BOTH — The Up and Down limit switches.

Field Service Level Categories and Parameters

Drill

Drill Idle Time Before Shutoff

Sets the time between drill commands before the drill shuts off. Use when the Drill On Only While Selected parameter is ON.

Drill on only while selected

ON — The drill is only on when a drill command is encountered in the cut data.

OFF — The drill stays on as long as the knife switch is on.

Servo (DSP) Coefficients

C Max Follow Error

Sets the maximum allowed amount of lag between the actual C-axis position and the commanded C-axis position without the system generating a following error.

Beam Bite Gain

Determines how closely the beam follows the conveyor during a conveyor advance. On certain systems, this parameter may need to be lowered to eliminate beam oscillation during the conveyor advance. Lowering this parameter causes a lag between the conveyor and beam position during the bite. This lag should not affect the accuracy of the bite, however, because the beam will catch up to the conveyor at the end of the bite.

Field Service Level Categories and Parameters

Time Considerations

Max time: To move up — bite sns

Sets the maximum time allowed for the transition of tools. If value is exceeded the system displays an error.

Not used on S-91 cutters.

Max time: To move down — bite sns

Sets the maximum time allowed for the transition of tools. If value is exceeded the system displays an error.

Always down on S-91 cutters.

Max time: To move up — drill 1

Sets the maximum time allowed for the transition of tools. If value is exceeded the system displays an error.

Max time: To move down — drill 1

Sets the maximum time allowed for the transition of tools. If value is exceeded the system displays an error.

Max time: To move up — knife

Sets the maximum time allowed for the transition of tools. If value is exceeded the system displays an error.

Max time: To move down — knife

Sets the maximum time allowed for the transition of tools. If value is exceeded the system displays an error.

Max time: To move up — pen

Sets the maximum time allowed for the transition of tools. If value is exceeded the system displays an error.

Max time: To move down — pen

Sets the maximum time allowed for the transition of tools. If value is exceeded the system displays an error.

Max time: Up to speed — knife

Sets the maximum time allowed for motor to turn on before a load is applied.

Max time: Up to speed — drill

Sets the maximum time allowed for motor to turn on before a load is applied.

Max time: To turn off — drill

Sets the maximum time allowed for motor to turn off.

Max time: To turn on — vacuum

Sets the maximum time allowed for motor to turn on before a load is applied.

Max time: To turn on — drives

Sets the maximum time allowed for the X, Y, C, Knife, and drill motors to turn on before a load is applied.

Max time: Idle in TrimCut Mode

Sets the maximum time the TrimCut mode stays active when in the idle state.

Max time: Idle in ManBite Mode

Sets the maximum time the manual bite mode stays active when in the idle state.

Max Time: Vacuum Drop

Sets the maximum time allowed for the vacuum to drop before a bite. If exceeded, the system displays an error.

Used for S-91 ZVC and TCW cutters only.

Max Time for Auto Bite

Sets the maximum time allowed for the conveyor to ramp down to a stop when the system turns the conveyor off during an Automatic Bite.

Not used on S-91 cutters.

Max Time: Engage Edge Sensor

Set the maximum time allowed for the edge sensor to engage the spread while automatically calculating Origin and Fabric Alignment for the Automatic Tube Centering System.

Field Service Level Categories and Parameters

Conveyor and CTOT

Velocity — Cnvyr Jog

Sets the speed of conveyor movement while jogging in the forward and reverse directions.

Not used on S-91 cutters.

Velocity — Cnvyr Man Bite

Sets the speed of conveyor movement while performing a manual bite.

Not used on S-91 cutters.

Velocity — Cnvyr Auto Bite

Sets the speed of conveyor movement while performing an automatic bite.

Not used on S-91 cutters.

Conveyor Ramp Down Time

Sets the time the system allows for the conveyor to ramp down to a stop when it turns off the conveyor after a jog or manual bite.

Not used on S-91 cutters.

CTOT — Stop Distance

Sets the distance from the end of the bite when the Conveyorized Take Off Table is shut off.

Not used on S-91 cutters.

Max Allowable Conveyor Friction

Sets the maximum amount of current in Amps needed to move the conveyor. (Used only by the Conveyor Friction Diagnostic; not yet implemented)

Not used on S-91 cutters.

TONT — Start Delay

Sets the time interval between when the conveyor starts moving and the conveyorized Take ON Table start signal is output.

TONT — Stop Distance

Sets the distance from the end of the conveyor advance when the conveyorized Take ON Table stop signal is output.

CTOT2 — Start Delay

Sets the time interval between when the conveyor starts moving and the Conveyorized Take Off Table start signal is output.

CTOT2 — Stop Distance

Sets the distance from the end of the conveyor advance when the Conveyorized Take Off Table stop signal is output.

Bite Encoder Step (beam)

Sets the size of one step as read by the bite encoder (located on the beam or head).

Conveyor Encoder LOC (0=B 1=T)

Identifies where the bite encoder is located on the cutter. Choices are:

BEAM or TABLE

Bite Encoder Step (table)

Sets the size of one step as read by the bite encoder (if located at the table).

Velocity — Cnvyr Cut Bite

Sets the speed at which the conveyor moves when the C-200MT program is biting a marker while cutting.

Variable Speed CTOT

ON — Tells the system that the type of CTOT for the cutter is variable speed.

OFF -Tells the system that the type of CTOT for the cutter is fixed speed.

Field Service Level Categories and Parameters

Software Parameters

Event Logging

ENABLE — Tracks events, such as cutter response to commands and messages displayed. Leave this feature turned on in order to use its tracking ability when recovering from error conditions.

DISABLE — Turns off this feature.

Note: This parameter in the Software Parameters category is set at DISABLE by default. To have the C-200MT program log information about Origin in Restore State, Opstop, and CIDB, Event Logging must be enabled.

Data Logging Mask

Identifies which motor response data is logged.

Milliseconds between samples

Sets the time between samples when data is being logged.

Log Writes to VARSAVE

ENABLE — Tracks events, such as cutter response to commands and messages displayed. Leave this feature turned on in order to use its tracking ability when recovering from error conditions.

DISABLE — Turns off this feature.

Note: To have the C-200MT program log information for this parameter, the Event Logging parameter must be turned ON.

Log OPSTOP Debug Info

ON — Tracks events that occur while working with Opstops using the AccuMatch system.

OFF — Turns off this feature.

Note: To have the C-200MT program log information for this parameter, the Event Logging parameter must be turned ON.

Log Origin Debug Info

ON — Tracks events that occur while finding the origin using the Automatic Tube Centering System.

OFF — Turns off this feature.

Log CIDB Debug Info

ENABLE — Tracks events that occur while generating CIDB reports.

DISABLE — Turns off this feature.

Note: To have the C-200MT program log information for this parameter, the Event Logging parameter must be turned ON.

Index

A

AAMA Data Conversion	19
AccuMatch Installed	35
AccuMatch, use	29
Active Beams	9
Advance Before Plunge Distance	9
Advance Ignore Angle	10
Allow Piece Reorder	20
Auto Start Jobs in Queue	34
Auto-Scale Y	24

B

Bar Code Reader Com Port	36
Bar Code Reader Option	27
Beam Bite Gain	39
Beam Square Adjustment	31
Beam Squared	31
Beam Width	35
Bite Clip End Position	21
Bite Clip Start Position	21
Bite Encoder Step (beam)	41
Bite Encoder Step (table)	41
Bite Error Adjustment	31
Bite Overlap	22
Bite Parameters	20
Bite Park Location	21
Bite Park Offset	21
Bite Vacuum Level	12
Bite Veeroff	21
Block Delete	26
Bristl Cleanr (0=N, 1=Motr, 2=Pneum)	14
Bristle Cleaner Parameters	14
Bristle Cleaner Sweeps per Slat	14

C

C Home to System 0 Degrees	31
C Max Follow Error	39
Camera Extension Distance	29
Categories	5, 35
category	9, 11, 13, 15, 18, 20, 24, 28, 29, 30, 31

Index

C-axis	37
Acceleration	37
cut velocity	37
delay	37
Max Angle	10
max velocity	37
Settle Time	37
Sharpen Speed	31
Yaw Angle	10
Center Line Calibration	19
Center Line Origin Center Point	26
Center Line Origin Edge Points	27
Check Origin	25
Cleaner Head Retract/Plunge Time	14
Cleaner Head Sweep Time	14
Clear Origin on Cutfile Close	24
Clipping High Boundary	28
Clipping High Boundary (Top Mrk)	28
Clipping Low Boundary	28
Clipping Low Boundary (Top Mrk)	28
Clipping Window Option	28
Contiguous Cut	30
Continuous Cutting	22
Continuous Cutting Installed	23
Convey Last Bite Off Table	20
Conveyor and CTOT	41
Conveyor Encoder LOC (0=B 1=T)	41
Conveyor Ramp Down Time	41
Conveyor Table Settle Time	22
Corner Advance Distance	9
Corner Angle	33
Corner Miter Distance	10
Corner Overcut Distance	9
CTOT - Stop Distance	41
CTOT2 Start Delay	41
CTOT2 Stop Distance	41
cut acceleration, max	31
Cut Distance Limit	7
Cut File Preprocessor Parameters	33
Cut Small Pieces First	22
Cut Speed=Maximum Knife Speed	6
Cut/Maximum Knife Speed	6
Cut/Plot Mode	26
Cutfile Percentage Primary Beam	37
Cutter Info Database Option	27
Cutter Info Database Output	29
Cutter Type	35
Cutting Parameters	9

Index

D

Data Format	18
Data Format/Scaling and RSO	18
Data Logging Mask	42
data scale factor	18
Del Job From Queue After Cutting.....	34
Del Queue After Cutting.....	34
Delete Cut File After Cutting.....	26
Distance to Corner	33
Drill Idle Time Before Shutoff	39
Drill on only while selected	39
drill speed parameter.....	6
Drill Speed Parameters	6
Dry Haul Speeds	36
Dry Haul x Acceleration	36
Dry Haul/Minimum Knife Speed.....	6
Dry Run	26
Dry Run Tool is Light.....	24
Dual Beam	35
Dual Beam Cutter	9

E

Edge Clip Code.....	21
Edge Clipping Angle	33
Edge Finder Auto-Alignment	24
Edge Finders	
installed	24
log info	24
use	24
Energy Savings Mode.....	12
Event Logging.....	42

F

Field Service Level Categories and Parameters .	35
---	----

G

Gap to Ignore	33
---------------------	----

H

Heavy Duty Drill Speed.....	6
home, C to System 0 Degrees	31

I

Idle Knife Speed	6
Ignore Auto Origin (M70) Codes	13

Ignore Aux Drill (M44) Codes.....	13
Ignore Codes Parameters.....	13
Ignore Display Msg (M20) Codes.....	13
Ignore Feedrate (F) Codes.....	13
Ignore Label Cycle (M31) Codes.....	13
Ignore Lift & Plunge (M46) Codes.....	13
Ignore Main Drill (M43) Codes	13
Ignore Move Conveyor (M69) Codes	13
Ignore OV AP Off (M19) Codes.....	13
Ignore Overcut Off (M18) Codes.....	13
Ignore Pen (D1+ D2) Codes.....	13
Ignore Sharpen (M42) Codes	13
Ignore Slowdown Codes	14
Insert M23 Edge Clip Code.....	21
Installed Edge Finders.....	24
Intermarker Gap	30

J

job queue	
auto start	34
del after cutting.....	34

K

Kanji Character Set	25
KI	
Corner Disable Time	15
Deflection Angle	15
Error Detection	15
Fail Time	15
Gain	15
Max Turn Angle	15
On/Off.....	15
Option.....	15
KI Spring Factor.....	15
Knife and Drill Speeds Parameters	6
Knife and Path Intelligence Parameters	15
Knife Break Force.....	17
Knife Break Risk Point	17
Knife Spd Sensor	
Volts per KRPM	6
Knife Speed Checking.....	6
knife speed parameter	6
Knife Wear Limit	8
Knife Wear Type.....	7
Knife Width.....	33

Index

L

Language Support.....	25
Length of Reverse Segment.....	33
Lift & Plunge Corner Angle.....	9
Limit Switch Status - Edge Sensor.....	38
Limit Switches.....	38
Lm Switch Status - Bite Sensor.....	38
Lm Switch Status - Drill 1.....	38
Lm Switch Status - Drill 2.....	38
Lm Switch Status - Knife.....	38
Lm Switch Status - Pen.....	38
Lm Switch Status - Sharpen Arm.....	38
Log CIDB Debug Info.....	42
Log Edge Finder Info.....	24
Log OPSTOP Debug Info.....	42
Log Origin Debug Info.....	42
Log Writes to VARSAVE.....	42

M

M17 (0 Heel Cut) Advance.....	10
M19 (Slit Notch) Advance.....	10
M25 Slowdown Percentage.....	10
Marker End Park (0=None, 1=-Y, 2=+Y).....	21
Marker End Park Position X.....	21
Marker End Park Position Y.....	21
Marker Flip.....	25
Marker Flip Top.....	25
Marker Size	
Use Auto Calculate.....	19
Use M20 Code.....	19
Marker Stacking Reorder.....	25
Marker Stacking Reorder Channel.....	25
Max # Points Per Piece.....	33
Max Allowable Conveyor Friction.....	41
Max Area of Small Piece.....	22
Max Auto-Scale Y Change.....	24
Max Cut Acceleration.....	31
Max Distance with Knife Down.....	33
Max Extension +Y Edge Finder.....	32
Max Extension -Y Edge Finder.....	32
Max Length to Ignore.....	33
Max time	
Engage Edge Sensor.....	40
for Auto Bite.....	40
Idle in ManBite Mode.....	40
Idle in TrimCut Mode.....	40
To move down - bite sns.....	40

To move down - drill 1.....	40
To move down - knife.....	40
To move down - pen.....	40
To move up - bite sns.....	40
To move up - drill 1.....	40
To move up - knife.....	40
To move up pen.....	40
To turn off - drill.....	40
To turn on - drives.....	40
To turn on - vacuum.....	40
Up to speed - drill.....	40
Up to speed - knife.....	40
Vacuum Drop.....	40
Maximum Bite Adjustment.....	21
Maximum Bite Length.....	20
Maximum Cut Velocity.....	31
Maximum Dry Haul Velocity.....	36
Maximum Segment Length.....	10
MCC Communications Port.....	36
Message Stop.....	33
Metric Display.....	26
Milliseconds between samples.....	42
Minimum Bite Length.....	20
Minimum Size of First Bite.....	30
Minimum Web Cut Size.....	30
Mirror Image.....	28
Mirror Image and Clipping Window.....	28
Mirroring Option.....	28
Miter Angle Range.....	10

N

Num of Vac Zones Closed for Bite.....	12
Num sharpens until angle adjust.....	8
Number of Sharpen Cycles.....	8
Number of Sides to Sharpen.....	7
Number of Table Slats.....	14
Number of Vacuum Zones.....	12

O

Offset Bottom Edge Sensor Y.....	32
Offset Edge Sensors X.....	32
Offset in X - Drill 1.....	31
Offset in X - Drill 2.....	32
Offset in X - Knife.....	31
Offset in X - Pen.....	31
Offset in Y - Drill 1.....	31
Offset in Y - Drill 2.....	32

Index

Offset in Y - Knife	31
Offset in Y - Pen	32
Offset Top Edge Sensor Y	32
Offset X - Camera	32
Offset Y - Camera	32
One-Way Bristle Cleaning	14
Only Allow Cutting with KI On	15
OPSTOP	29
OPSTOP and Report Options	29
OPSTOP Section Processing	29
OPSTOP Slow Slew	29
OPSTOPS	29
Optimal Backup Distance	9
Options and Modes	24
Overcut Distance	9
Overlap Distance	33

P

parameters	5, 7, 9, 11, 13, 15, 18, 20, 24, 28, 29, 30, 31, 35
Parameters	6
park	21
bite offset	21
mark end	21
marker end position X	21
marker end position Y	21
Path Intelligence Option	16
Path Overlap	33
Pause Before Convey	20
Pause Between Bites	20
PC Serial Communication	36
Percent of Piece	33
PI	
Feedrate	
Change Ratio	16
Max Increase	16
Maximum	16
Minimum	16
Force to Stop with Warning	16
FR Control - XY Force	16
FR Faster - Shp Knife	16
FR Slowdown - Low Vac	16
Optimum Force	16
Sharp Knife Full Speedup Dst	16
Sharp Knife Prtl Speedup Dst	17
Sharp Knife Speedup Amount	17
Vacuum Low Slowdown Factor	17
Velocity to +FR	16

Velocity to +FR & Rbld	16
Velocity to -FR	16
Velocity to -FR & Rbld	16
Preprocess Drills	26
Preprocessor Parameters	33
Preset OPSTOP	29
Projector Output Port	36

Q

Queue Manager Parameters	34
--------------------------------	----

R

Ramp Down Dist - Auto Bite	32
Read Queue Name Bar Codes	34
Recalculate Small Pieces	22
Regional OPSTOP	29
Re-Sealer Option	26
Resealer Up For Continuous Cut	26
Right Side Control	35
Right Side Initialize	19
RSO	19, 35
Run Time Biting	20

S

Saving of Parameter Changes	24
Security Parameters	34
Selv/Web (0=Off, 1= Selv, 2 = Web&Sel)	30
Selvage Cut Position	30
Selvage Cut Size Bottom	30
Selvage Cut Size Top	30
Selvage Cut/Contiguous Cut	30
Servo (DSP) Coefficients	39
Sharpen Every n Pieces	7
Sharpen Frequency	7
Sharpen Knife Speed	6
Sharpen New Knife n Times	8
Sharpen Only When Knife is Up	7
Sharpen Parameters	7
Sharpen Position #1	7
Sharpen Position #2	7
sharpen speed, C-axis	31
Sharpening Time	7
Sharpens Until Overcut Update	8
Single Step	25
Slat Advance Time	14
slit notch	9

Index

Slit Notch Backtracking	9
small pieces	
cut first	22
max area	22
reacalculate	22
Smallest Piece	33
Software Parameters	42
System Setup Parameters.....	31

T

Table Length (0=Standard, 1=Long)	35
table, usable distance	31
Time Considerations	40
Time Until ESM	12
TONT Start Delay.....	41
TONT Stop Distance	41
Transfer Angle	33
Transfer Markers When Selected	27

U

Usable Table Distance	31
Use AccuMatch	29
Use Edge Finders	24
User Parameter Table	
editing	34
selecting	34

V

Vacuum Control - Pattern Length.....	12
Vacuum Control - Vacuum Gain	12
Vacuum Control - Zero Offset.....	12
Vacuum Control Type	11
Vacuum Error Reporting	11
Vacuum Level - Bite.....	11
Vacuum Level - Cut.....	11
Vacuum Level - Idle	12
Vacuum Level Valve on Inlet.....	12
Vacuum Low Caution Level.....	12
Vacuum Low Error Level	12
vacuum parameter.....	12
Vacuum System	11
Vacuum Zone Overlap.....	11
Vacuum Zoning	11
Variable Length Biting	20
Variable Speed CTOT	41
Velocity - Cnvyr Auto Bite.....	41

Velocity - Cnvyr Jog	41
Velocity - Cnvyr Man Bite.....	41
Velocity Cnvyr Cut Bite.....	41
Verify Keyboard Entry.....	26

W

web, minimum cut size	30
web, selv/	30

X

X Data Scale Factor	18
X Home to System 0	37
X Home Zone1 Offset.....	12
X-axis	
Stepsize.....	31
XY Axes.....	37
XY-Axes delay.....	37

Y

Y Data Scale Factor	18
Y Home to System 0	37
Y-axis	
Edge Align (0=N, 1=+Y, 2=-Y)	27
Mirror Image	28
Mirror Reflection Line	28
Stepsize.....	31

Index